Subject: Mathematics | **PERIODIC TEST – 04 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Show clear steps in all probability derivations and calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. Three identical unbiased dice are rolled together. The probability that the same number appears on all three dice is: **[1]**

Options: (a) $\frac{1}{36}$ (b) $\frac{1}{216}$ (c) $\frac{1}{6}$ (d) $\frac{5}{36}$

2. From the numbers $3, 5, 5, 7, 7, 7, 9, 9, 9, 9$, one number is selected at random. The probability that the selected number is their mean is: **[1]**

Options: (a) $\frac{3}{10}$ (b) $\frac{1}{10}$ (c) $\frac{7}{10}$ (d) $\frac{2}{5}$

3. A lot of 20 bulbs contains 4 defective ones. One bulb is drawn at random from the lot. The probability that this bulb is non-defective is: **[1]**

Options: (a) $\frac{4}{5}$ (b) $\frac{1}{5}$ (c) $\frac{3}{4}$ (d) $\frac{1}{4}$

4. The probability of guessing the correct answer to a certain test question is $\frac{x}{12}$. If the probability of not guessing the correct answer is $\frac{2}{3}$, then x is equal to: **[1]**

Options: (a) 2 (b) 3 (c) 4 (d) 6

5. **Assertion (A):** The probability that a non-leap year has 53 Tuesdays is $\frac{2}{7}$.
Reason (R): A non-leap year contains 365 days, giving 52 weeks and 1 odd day. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. A bag contains 4 red, 3 blue, and 2 yellow balls. One ball is drawn at random. Find the probability that the ball drawn is: (i) red or blue (ii) neither yellow nor red. **[2]**

7. The probability that it will rain tomorrow is 0.85. What is the probability that it will not rain tomorrow? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. A piggy bank contains a hundred 50p coins, fifty Rs. 1 coins, twenty Rs. 2 coins, and ten Rs. 5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin: (i) will be a 50p coin? (ii) will not be a Rs. 5 coin? (iii) will be either a Rs. 1 or a Rs. 2 coin? **[3]**

9. Two dice are thrown at the same time. Find the probability that: (i) the difference of the numbers on the two dice is 2 (ii) the difference of the numbers on the two dice is 0 (iii) the number on the first die is greater than 4. **[3]**

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

10. A box contains 12 balls out of which x are black. (i) If one ball is drawn at random from the box, what is the probability that it will be a black ball? (ii) If 6 more black balls are put in the box, the probability of drawing a black ball is now double of what it was before. Find the value of x. (iii) With this value of x, find the probability of drawing a non-black ball from the final box. (iv) If 2 white balls are now removed from the box, find the new probability of drawing a black ball. **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 11: Case Study Analysis

Mira designs a target game using a circular cardboard of radius $R = 14$ cm with center O. The cardboard is divided into three distinct concentric regions:

- **Region X (Inner Bullseye):** A central circle of radius $r_1 = 7$ cm.
- **Region Y (Middle Ring):** An annular ring between radius $r_1 = 7$ cm and $r_2 = 14$ cm.
- **Region Z (Semicircular Sector):** A half-annulus formed by a diameter AB cutting across Region Y. A pellet is dropped at random onto the circular board. Taking $\pi = \frac{22}{7}$:

Based on the above scenario, answer the following:

(i) Find the total area of the complete circular cardboard disc. **[1]**

(ii) Find the area of Region X (the inner bullseye circle). **[1]**

(iii) What is the probability that the pellet lands in Region X? **[1]**

(iv) What is the probability that the pellet lands in Region Y (the annular ring)? **[1]**

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